

Energy-system interactions reshape offshore hydrogen trade and carbon mitigation in Europe

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Abstract

Offshore wind-rich island systems are often assessed through production costs, but their hydrogen export value also depends on domestic absorption, shipping, demand geography and competing imports. Here we present the first integrated European-scale modelling framework linking offshore hydrogen cost-supply curves, hourly power-gas operation, cross-border hydrogen/ammonia shipping and European demand from 2030 to 2050. A lowest-LCOH-only allocation identifies Denmark or the Netherlands as leading exporters, whereas the integrated model identifies the UK as the leading net exporter in all three model years, with Ireland nearly matching it by 2050. In the integrated 2050 case, the UK and Ireland each reach approximately 162 TWh yr⁻¹ of net exports, and modelled offshore hydrogen contributes about 175.5 Mt CO₂ yr⁻¹ of emissions mitigation. The result is conditional on the external import option: delivered outside prices near 2–3 EUR kg⁻¹ H₂-equivalent displace most European offshore supply, whereas prices around and above 4 EUR kg⁻¹ restore the UK/Ireland export pattern. These findings show that strategic offshore hydrogen exporters cannot be inferred from LCOH alone.

Keywords: hydrogen, offshore wind, carbon mitigation, international trade

1 Introduction

Europe’s offshore wind expansion creates a spatial allocation problem, not only a generation-cost problem. Ireland and the UK have announced offshore wind development plans reaching 37 and 125 GW of installed capacity, respectively, by 2050 [1, 2]. These targets are more than twice the projected values of their total power demands in 2050 [3, 4]. Connecting and using such large offshore generation volumes through domestic power systems alone is challenging because domestic demand is limited and real-time balancing requirements increase with uncertain and intermittent offshore wind.

Green hydrogen offers a route for moving offshore renewable energy across sectors and borders. Renewable electrolysis avoids the direct operational emissions associated with grey hydrogen and can reduce lifecycle emissions relative to fossil-based production routes [5]. Hydrogen can also be stored and transported at larger scales than electricity, allowing offshore wind output to serve domestic industry, power and heat demands or distant demand centres through pipelines and shipping. This matters because prospective exporters such as Ireland and the UK sit near large future import needs: Germany expects annual imported hydrogen needs of 45–77 TWh by 2030 [6, 7], and Belgium has identified 20 TWh of imported hydrogen demand by 2030 [8].

Production cost remains central to offshore hydrogen deployment, but it is not sufficient for identifying strategic exporters. The levelised cost of hydrogen (LCOH) in 2030 has been projected at around 54–184 €/MWh, overlapping with blue hydrogen with carbon capture and storage and green hydrogen from onshore wind [9]. Country- and region-specific LCOH studies have assessed North Sea countries [10], the UK [11], Poland [12] and the Iberian Peninsula [13, 14], often with detailed cost breakdowns, production routes and uncertainty analysis. However, separate national studies cannot show how production costs interact with domestic absorption, shipping costs, European demand geography and competing imports.

This gap is particularly important for Ireland and the UK. European hydrogen strategies often emphasise imports from North Sea countries, North Africa, the Middle East and the Americas, while the export potential of these Atlantic-facing island systems is less quantified. Policy and consulting reports have identified opportunities for Irish and UK hydrogen exports [1, 15], but a pipeline-only or LCOH-only view can miss how their offshore resource value changes once domestic energy-system constraints and European trade balances are considered together. A feature comparison with representative prior study types is provided in Supplementary Table 1.

This work provides the first integrated European-scale assessment of offshore hydrogen trade that links geospatial production costs, hourly domestic power-gas absorption, shipping-based hydrogen and ammonia exchange, and competing external imports in one optimisation framework. We first construct geospatial cost-supply curves for offshore hydrogen in 11 coastal countries using meteorological, bathymetric, infrastructure and exclusion-area data. We then use an hourly coordinated power-gas optimisation model for Ireland to estimate how domestic energy systems absorb offshore wind and hydrogen under evolving operating constraints, and extend the resulting export-potential logic to the wider country set. Finally, we combine cost-supply curves, domestic export limits, shipping costs, national hydrogen and

ammonia demand, and an external import option in an international trade optimisation model. This structure lets us test whether strategic exporters can be inferred from LCOH alone and how robust the resulting trade geography is to competing imported hydrogen.

2 Results

2.1 Offshore hydrogen costs do not determine export value alone

We assess Belgium, Denmark, France, Germany, Ireland, the Netherlands, Norway, Portugal, Spain, Sweden and the United Kingdom (UK), which combine large exclusive economic zones (EEZs), offshore infrastructure potential and navigable links to European demand centres. Wind velocity, air pressure and temperature data are obtained from ERA5 at 1 hour and $0.5^\circ \times 0.5^\circ$ temporal and spatial resolutions [16]. Hourly capacity factors and wake-effect losses are simulated using a 15 MW reference wind turbine [17]. The resulting wind-speed distributions show that Ireland has the highest modal wind speed among the studied EEZs, followed by the UK, mainly because of their exposure to Atlantic wind resources (Fig. 1b).

We then combine these capacity factors with development, capital, operating and decommissioning costs for offshore hydrogen production platforms, including turbines, foundations, electrolyzers and auxiliary systems (Methods). Bathymetry, port locations and power/gas connection points are included so that water depth, installation distance and maintenance distance affect site-level costs [18, 19]. The resulting LCOH ranges from 84.93 to more than 200 €/MWh across the studied EEZs (Fig. 1a). Denmark has the lowest-cost locations because it combines high capacity factors with a shallow continental shelf, although the lower-bound offshore hydrogen cost remains 33.94% higher than an onshore blue-hydrogen benchmark of 68.92 €/MWh [20]. The UK’s LCOH ranges from 85.97 to 307.26 €/MWh, while Ireland’s ranges from 90.42 to 201.8 €/MWh. These results show that Atlantic wind-resource quality does not automatically translate into the lowest production cost once foundation depth and infrastructure distances are included.

We further account for wake losses, vessel density, excluded areas such as military and aquaculture zones, and future cost reductions [21] to construct country-level cost-supply curves for 2030, 2040 and 2050 (Fig. 2). In 2030, marginal LCOH is 76.10 €/MWh in Ireland and 79.02 €/MWh in the UK (Fig. 3). Ireland is slightly cheaper at the policy-constrained marginal capacity, but the UK remains more competitive over several equal-volume comparisons because it has more low-cost near-shore supply. Denmark is the leading cost benchmark, while France and the Netherlands retain competitive supply segments even when their full offshore capacity is not the lowest-cost option.

Technology and supply-chain learning reduce LCOH in later scenarios, but the country ranking remains capacity-dependent. In 2040, Ireland’s marginal LCOH is 67.40 €/MWh because its larger offshore target pushes production into higher-cost sites. The UK improves relative to Sweden because of favourable capacity factors, water depths and coastal access, while Denmark remains the leading cost-competitive

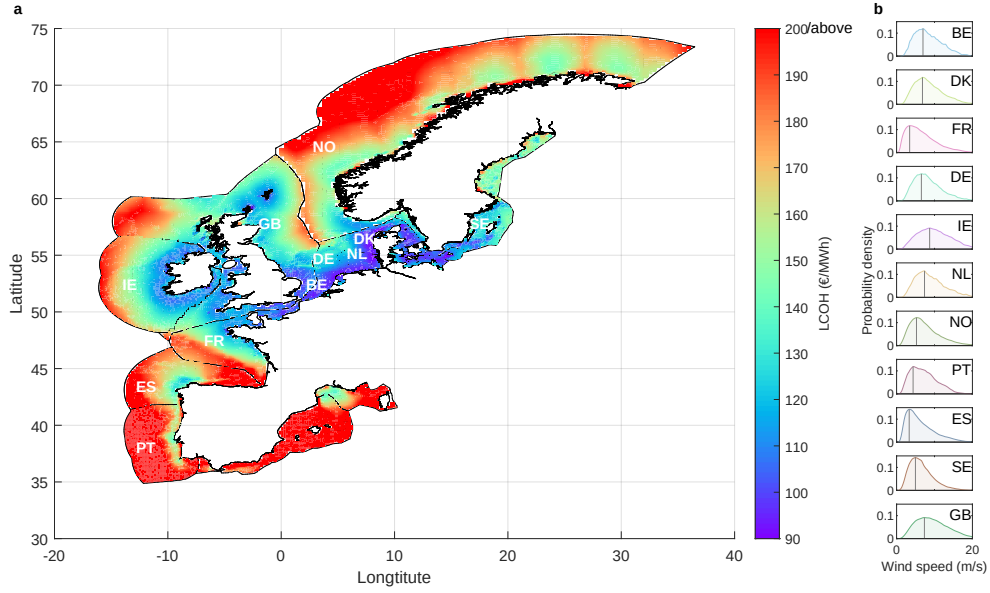


Fig. 1 LCOH map and wind speed in coastal European countries. **a** LCOH map in EEZs of coastal European countries. The black line outlines the EEZs. **b** Probability density of wind speed for each country. The vertical grey line in each figure shows the wind speed most likely to occur. BE, DK, FR, DE, IE, NL, NO, PT, ES, SE, and GB are the abbreviations for Belgium, Denmark, France, Germany, Ireland, The Netherlands, Norway, Portugal, Spain, Sweden, and the United Kingdom, respectively.

country. By 2050, marginal LCOH falls to 54.42 €/MWh for Ireland and 52.51 €/MWh for the UK, and all studied countries have offshore hydrogen segments that can compete with the assumed blue-hydrogen cost range. Overall, Denmark dominates the LCOH ranking, whereas Ireland and the UK occupy mid-ranking cost positions. Their strategic value therefore cannot be established from LCOH alone and must be tested against domestic absorption, shipping and demand.

2.2 Domestic power-gas constraints create exportable offshore surplus

Because the offshore wind ambitions of Ireland and the UK exceed their projected domestic power demands, we next estimate how much offshore production can be absorbed domestically before it becomes available for export. We develop a whole-energy-system optimisation model that simulates hourly power and gas operation over a full year under uncertain offshore wind generation. Power and natural gas systems are coupled through gas-fired power plants and onshore/offshore electrolyzers, allowing bidirectional energy conversion to expand system flexibility. Hydrogen blending and pure-hydrogen network cases are represented through different maximum hydrogen fractions in the gas network. Hourly unit commitment includes the

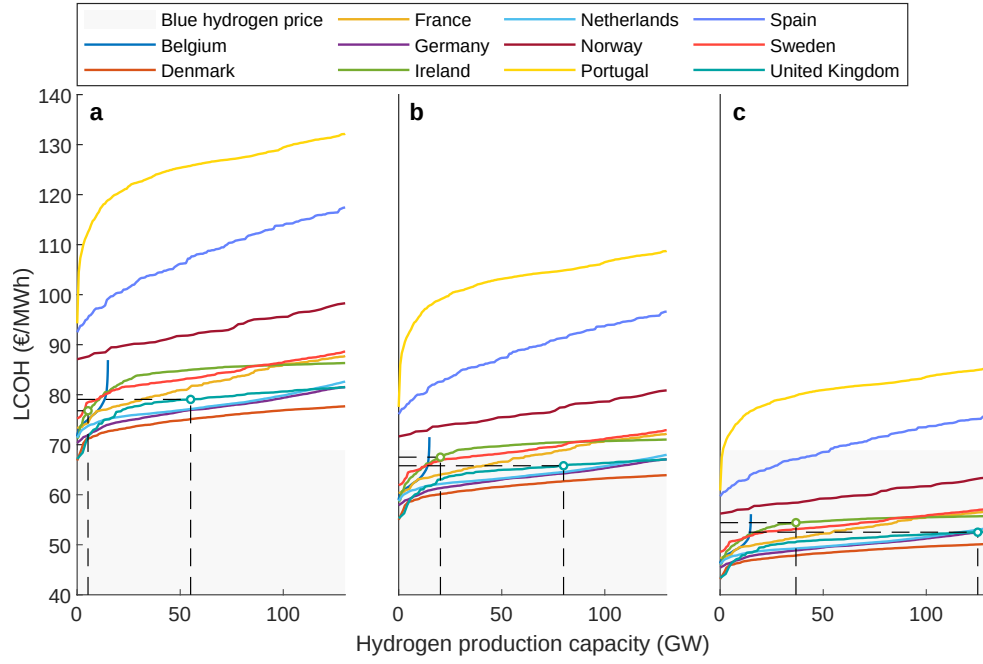


Fig. 2 Hydrogen cost-supply curves for different countries at different years. The cost-supply curves of green hydrogen produced by countries in 2030, 2040, and 2050 are shown in **a**, **b**, and **c**, respectively. Different colours of solid lines represent different countries. Green and blue dots on Ireland's and the UK's curves mark their goals for offshore wind capacities. Grey areas represent the possible range of blue hydrogen production prices in the future.

All-Ireland generator fleet, interconnectors, reserve, inertia, must-run and system non-synchronous penetration constraints, and observed hourly demand profiles (Methods; Supplementary Information) [22].

The domestic absorption summary compares seasons, model years and export/no-export cases (Fig. 4). We assume a maximum 20% hydrogen blending ratio in 2040, reflecting near-term blending demonstrations and policy movement, and a pure-hydrogen network in 2050 in line with the UK's hydrogen network roadmap [23]. Without export, the average summer wind-curtailment rate increases from 62.36% in 2030 to 69.80% in 2050 because high offshore generation coincides with low seasonal demand. In winter, the curtailment rate falls from 32.05% in 2030 to 13.47% in 2050 because higher hydrogen blending allows the gas system to absorb more offshore energy.

Even after extensive hydrogen use in the gas system, domestic demand cannot absorb the full offshore wind output. Export access substantially increases utilisation across all model years and seasons by creating an outlet for surplus hydrogen. In the 2050 summer case, for example, the average wind-curtailment rate falls to 10.46% with export access, and the full-year decomposition identifies approximately 170 TWh yr^{-1} of Irish export potential. Curtailment does not fall to zero because operating reserves,

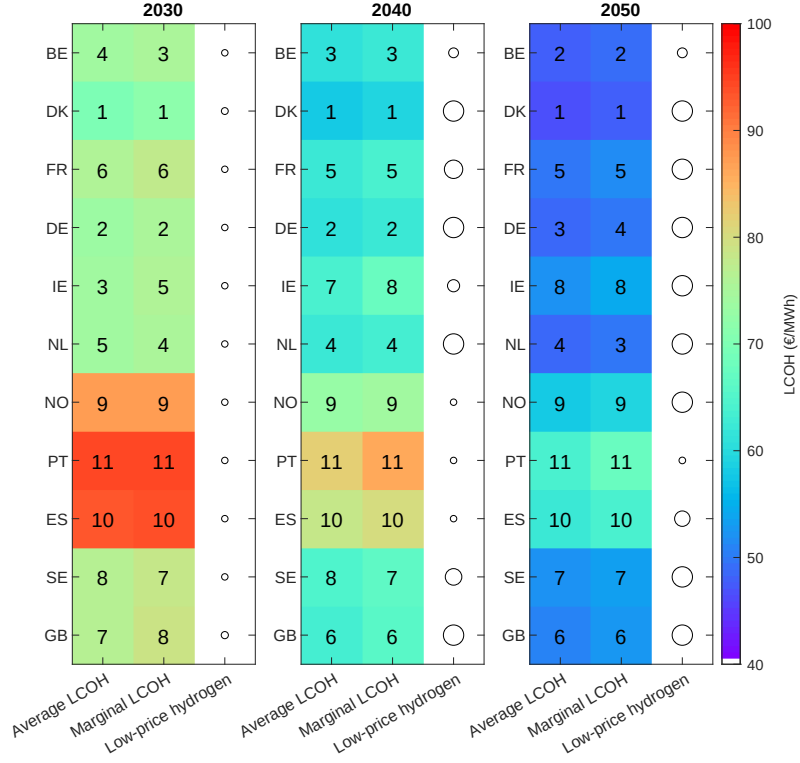


Fig. 3 Cost-competitiveness of offshore green hydrogen among major countries and with blue hydrogen. Rank indicates the average or marginal LCOH rank among the 11 studied countries. Marginal means the LCOH at the policy-constrained offshore capacity of that country in the corresponding year. Average means the capacity-weighted average LCOH across the offshore capacity in the corresponding year. Low-price hydrogen means the hydrogen production capacity with production cost below the upper bound of blue hydrogen. The size of the circle indicates the capacity of low-priced hydrogen. The smallest and largest circle sizes indicate 0 and >100 GW, and intermediate sizes indicate capacities between these values.

inertia requirements and must-run constraints still limit how much uncertain offshore output can be used at the margin.

We then use the full-year Ireland results to estimate offshore wind utilisation for the wider country set, assuming that domestic offshore absorption scales with electricity and gas demand. The annual decomposition identifies available export potential rather than optimised net trade (Fig. 5). In 2030, the UK has the largest available export potential at 176 TWh yr⁻¹ because its offshore capacity is large relative to domestic load. The Netherlands, Denmark and Ireland follow with 64, 30 and 16 TWh yr⁻¹, respectively. In 2040, the UK and the Netherlands remain the two largest potential exporters, with 195 and 155 TWh yr⁻¹, while Ireland reaches 87 TWh yr⁻¹ and becomes the third-largest potential exporter. By 2050, higher hydrogen use in gas networks increases domestic absorption in some countries, so the Netherlands' available export potential decreases by 13.6% even though offshore capacity increases. In contrast, the UK and Ireland continue to increase by 34.8% and 95.0% relative to

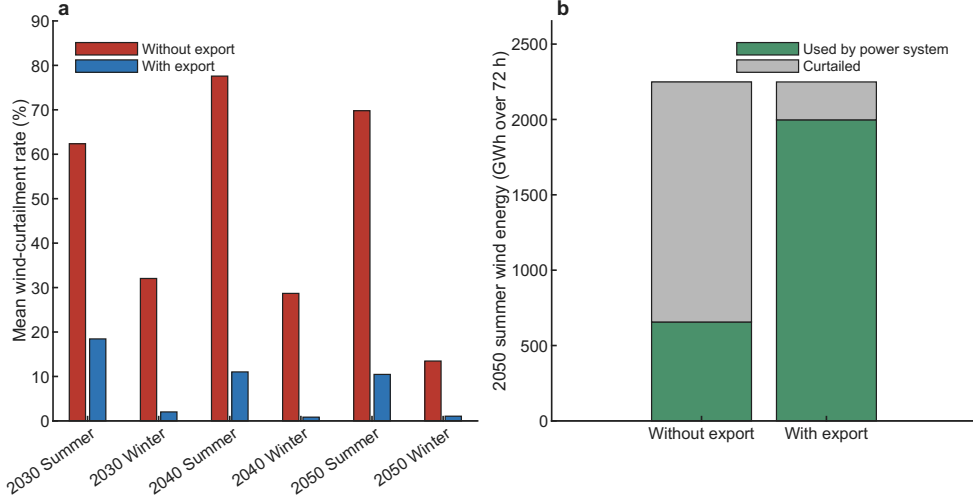


Fig. 4 Export access reduces domestic wind curtailment in the Ireland power-gas case. **a** Mean wind-curtailment rates across representative summer and winter periods in 2030, 2040 and 2050, with and without export access. **b** Decomposition of available wind energy in the 2050 summer representative period. Export access converts otherwise curtailed wind into energy used by the power system, but residual curtailment remains because operating constraints still bind at the margin.

2040, reaching 263 and 170 TWh yr⁻¹, respectively. Ireland therefore becomes the second-largest source of available export potential, despite not being a lowest-LCOH country.

2.3 Production cost alone misidentifies strategic hydrogen exporters

The cost and domestic-absorption results above show that low LCOH is only one part of the offshore hydrogen trade problem. To isolate what is missed by a production-cost view, we constructed a lowest-LCOH-only counterfactual that allocates hydrogen demand to the lowest-cost offshore supply first, and compared it with the integrated model that includes domestic absorption limits, shipping costs, national demand and residual external imports. This counterfactual is not intended as a market-design proposal; it is a diagnostic baseline for testing whether the strategic-exporter ranking can be inferred from LCOH alone.

The two approaches give different exporter geographies (Fig. 6). In the cost-only baseline, the leading net exporters are Denmark in 2030, the Netherlands in 2040 and Denmark again in 2050, with net exports of 44, 147 and 171 TWh yr⁻¹, respectively. In the integrated model, the leading net exporter is the UK in all three model years, with 78, 122 and 162 TWh yr⁻¹, while Ireland nearly matches the UK in 2050 at 162 TWh yr⁻¹. The difference is particularly large in 2050: the cost-only baseline assigns 41 TWh yr⁻¹ of net exports to Ireland and a 44 TWh yr⁻¹ net import position to the UK, whereas the integrated model increases both countries to approximately 162 TWh yr⁻¹ of net exports. The integrated model also reveals residual external import requirements of approximately 116 and 62 TWh yr⁻¹ in 2040 and 2050, while the

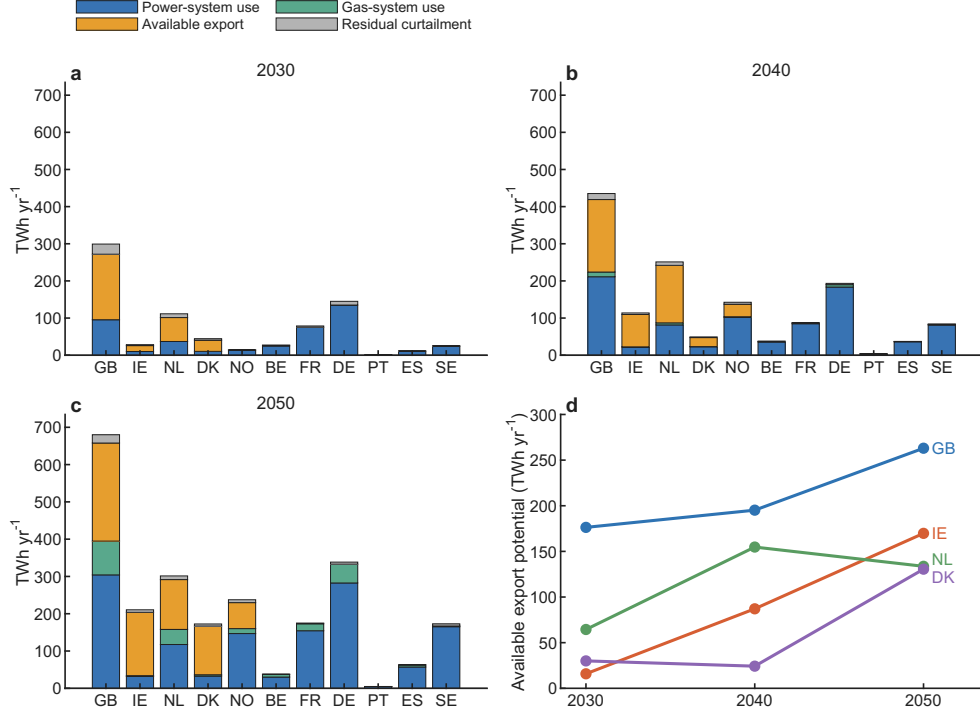


Fig. 5 Domestic absorption creates uneven exportable offshore surplus across Europe. **a–c** Country-level decomposition of offshore wind use in 2030, 2040 and 2050, ordered by 2050 available export potential. Power-system use, gas-system use, available export potential and residual curtailment are reported in TWh yr^{-1} . **d** Export-potential trajectories for the United Kingdom, Ireland, the Netherlands and Denmark.

cost-only baseline hides this shortage by construction. These results show that the policy-relevant exporter ranking is determined by the system balance between cost, domestic absorption, shipping and demand, rather than by LCOH ranking alone.

2.4 Integrated trade reallocates carbon-mitigation value across Europe

We next combine the cost-supply curves and export-potential estimates in an international green hydrogen trading model for 2030, 2040 and 2050. The cost-supply curves are interpolated as piecewise linear functions and combined with shipping costs, national hydrogen and ammonia demand, domestic export limits and external imports in a cost-minimising optimisation. The resulting flows and supplier contributions to carbon mitigation are shown in Fig. 7.

In 2030, the UK is the major hydrogen exporter, followed by the Netherlands, Denmark, and Ireland, with net exports of approximately 78, 39, 27 and 14 TWh yr^{-1} , respectively. Germany remains the major hydrogen receiver due to its high demand, so the lowest-cost and geographically close suppliers do not determine the whole trade pattern by themselves. By 2040, external hydrogen imports become necessary because

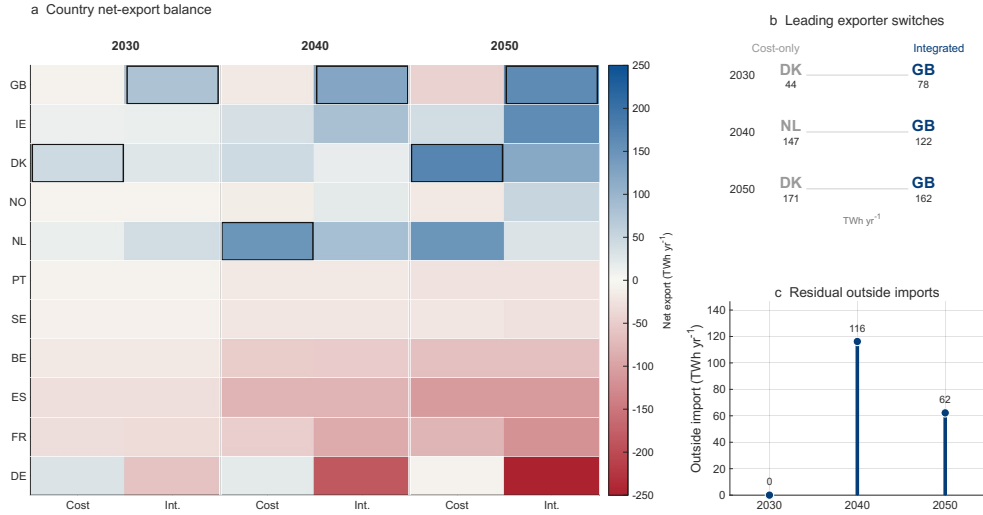


Fig. 6 Lowest-LCOH allocation misidentifies strategic offshore hydrogen exporters. **a** Country net-export balance under a lowest-LCOH-only baseline and the integrated model in 2030, 2040 and 2050. Blue indicates net export and red indicates net import. Black boxes mark the leading net exporter in each model-year combination. **b** Leading net exporter and corresponding net export under the two model structures. **c** Residual outside imports identified by the integrated model. The cost-only baseline allocates demand within the modelled European offshore supply by construction and therefore does not expose the residual external import requirement.

offshore capacity growth is slower than demand growth. The integrated solution identifies approximately 116 TWh yr⁻¹ of external imports, while the UK, the Netherlands and Ireland remain large European net exporters with approximately 122, 85 and 83 TWh yr⁻¹, respectively. In 2050, external imports fall to approximately 62 TWh yr⁻¹ but remain part of the overall supply structure. The UK and Ireland become the two largest European net exporters at approximately 162 TWh yr⁻¹ each, followed by Denmark at approximately 118 TWh yr⁻¹.

Hydrogen trade also reallocates carbon-mitigation value across Europe (Fig. 7b,d,f). Offshore hydrogen production from these 11 countries contributes approximately 43.5, 129.8 and 175.5 Mt CO₂ yr⁻¹ of emissions mitigation in 2030, 2040 and 2050, respectively, equivalent to 0.74, 2.20 and 2.98 times Ireland's current annual carbon emissions [24]. In 2030, the UK and the Netherlands are the main contributors to carbon mitigation. From 2040 to 2050, Ireland's importance increases with its export position. Ireland and the UK therefore mitigate emissions not only through domestic supply, but also through hydrogen delivered to demand centres elsewhere in Europe.

2.5 External import prices define a threshold for European offshore supply

The integrated trade result depends on how attractive non-European low-carbon hydrogen or ammonia imports become relative to European offshore supply. This outside option is strategically important because European policy already anticipates substantial renewable hydrogen imports, while the future scale, timing and cost of

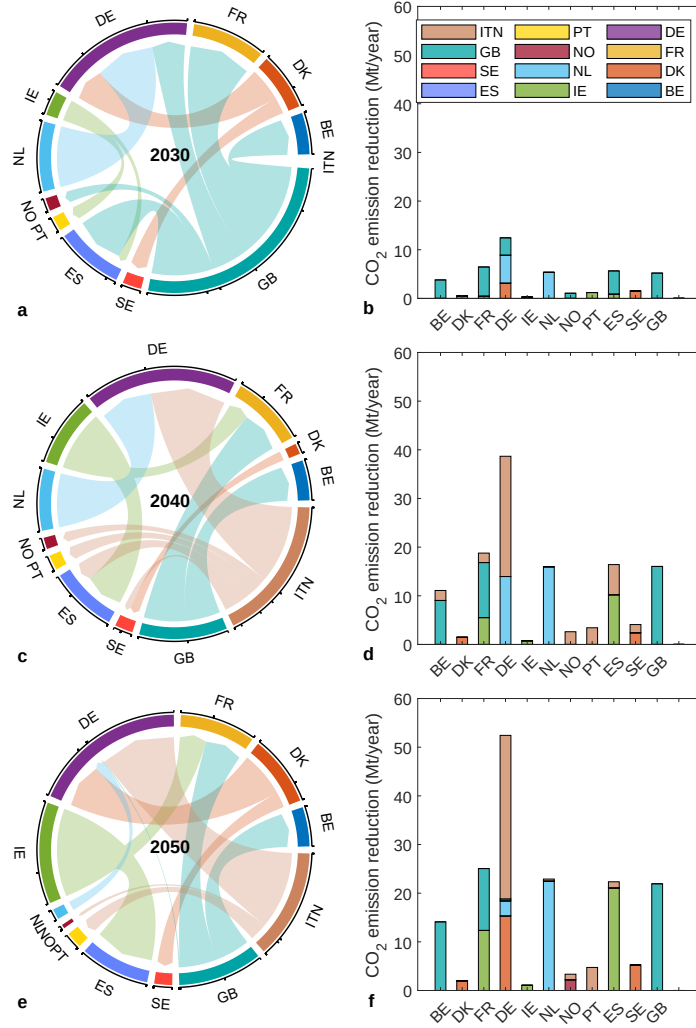


Fig. 7 Sankey diagram of hydrogen flows and contributions to carbon mitigation. a, c, and e, are the Sankey diagrams of hydrogen flows among these countries and external hydrogen imports in 2030, 2040, and 2050, respectively. ITN represents external import. Only the hydrogen flows between countries are drawn; hydrogen supplied and consumed within the same country is not shown. b, d, and f show each exporting country's contribution to carbon mitigation through international green hydrogen trading. Different bars represent the reduction in carbon dioxide emissions, and different colours indicate different countries' contributions.

global low-emission hydrogen trade remain uncertain [25–27]. We therefore replaced the previous high import penalty with a delivered hydrogen-equivalent outside-option price and solved the trade model over 2–6 EUR kg^{−1} H₂-equivalent. This range should be read as a threshold scan rather than a price forecast: its low end is informed by optimistic global trade and cost estimates, while the upper range represents less favourable delivery, infrastructure, finance or policy conditions [28, 29].

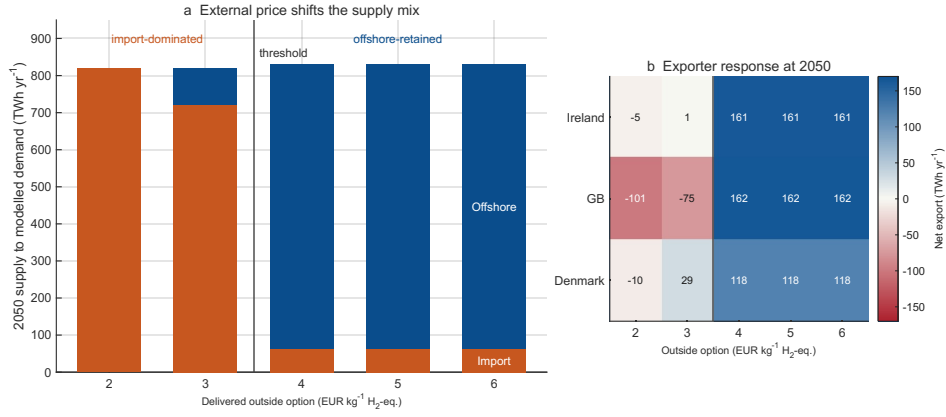


Fig. 8 External import prices define a threshold for European offshore hydrogen supply. **a** 2050 supply mix between external low-carbon imports and European offshore hydrogen under delivered outside-option prices from 2 to 6 EUR kg⁻¹ H₂-equivalent. **b** Net-export response of Ireland, the United Kingdom and Denmark under the same price scan. Blue indicates net export and red indicates net import. The scan is designed to identify a price threshold, not to assign probabilities to future import prices.

The outside-option scan shows a threshold rather than unconditional robustness (Fig. 8). In the 2050 case, an external option near 2 EUR kg⁻¹ H₂-equivalent displaces European offshore supply almost entirely, with 818 TWh yr⁻¹ supplied from outside the modelled European offshore system. At 3 EUR kg⁻¹, European offshore supply remains limited to 98 TWh yr⁻¹, outside imports remain at approximately 721 TWh yr⁻¹ and the UK remains a net importer. Around 4 EUR kg⁻¹, the solution changes regime: European offshore supply rises to about 767 TWh yr⁻¹, outside imports fall to about 63 TWh yr⁻¹, and the UK, Ireland and Denmark re-emerge as large net exporters, at approximately 162, 161 and 118 TWh yr⁻¹, respectively. The UK/Ireland export conclusion is therefore conditional on external imports not being available at ultra-low delivered prices, but it is restored once the outside option enters a more constrained delivered-cost range.

3 Discussion

The overall analysis shows that offshore green hydrogen can contribute materially to Europe’s net-zero transition, but its strategic geography is not determined by LCOH alone. Denmark remains the strongest cost-competitiveness benchmark across the cost-supply curves, whereas Ireland and the UK become strategically important only after domestic absorption, national demand, shipping and residual external imports are considered together. In the integrated 2050 case, the UK and Ireland become the two largest net exporters with approximately 162 TWh yr⁻¹ each, followed by Denmark with approximately 118 TWh yr⁻¹.

This distinction matters for policy appraisal. A lowest-cost-only allocation would identify Denmark or the Netherlands as the leading exporters and would hide residual

import dependence, whereas the integrated model identifies a west-to-east trade structure with persistent external import needs in 2040 and 2050. The result is therefore not that high-resource island systems automatically dominate offshore hydrogen trade. Rather, their strategic role emerges when large offshore build-out coincides with limited domestic absorption, nearby continental demand and a delivered external import option that is not ultra-low cost.

These findings also clarify how Ireland should be interpreted in future hydrogen planning. Ireland has no decisive LCOH advantage and does not have the largest offshore hydrogen production capacity. Its strategic value arises from the combination of sufficient cost competitiveness, large exportable surplus and European demand centres that remain undersupplied by their domestic offshore resources. This conclusion supports considering multiple export corridors and carrier options rather than evaluating Irish exports only through a single pipeline route. It also implies that national offshore hydrogen strategies should be assessed within a regional trade and demand balance, not only through domestic cost-supply estimates.

The central innovation of this study is therefore methodological as well as quantitative. By linking geospatial offshore hydrogen cost-supply curves, hourly domestic power-gas absorption, hydrogen/ammonia shipping and an explicit external import option, the model exposes a ranking reversal that single-module assessments cannot capture. The lowest-LCOH baseline points to Denmark or the Netherlands as leading exporters, but the integrated model identifies the UK as the leading exporter in all three model years and Ireland as an almost equal exporter by 2050. This first integrated European-scale treatment is what makes the UK/Ireland result visible.

Several assumptions define the scope of these conclusions. The analysis uses currently published offshore wind goals and therefore assumes that planned infrastructure is delivered on schedule. Domestic absorption outside Ireland is estimated by scaling the Ireland whole-system result with national electricity and gas demand, so the export-potential estimates should be read as a European screening result rather than a replacement for country-specific unit-commitment studies. A $\pm 25\%$ domestic-absorption sensitivity reported in the Supplementary Information changes the exact UK/Ireland ordering in 2050, but Ireland remains a leading net exporter and the LCOH-only ranking reversal is preserved. The outside-option analysis is also a threshold scan, not a forecast of future global hydrogen prices. These assumptions do not weaken the integrated trade result, but they identify where country-level planning and future market evidence would most improve the analysis.

The results indicate that residual external imports remain necessary in the integrated 2040 and 2050 cases, at approximately 116 and 62 TWh yr^{-1} , respectively. More ambitious offshore hydrogen production goals could reduce this dependence, especially in countries with lower marginal costs and remaining offshore capacity potential, such as Denmark and the Netherlands. Infrastructure policy also matters: investment in offshore wind turbines, electrolyzers, subsea cables, ports and hydrogen shipping vessels must be coordinated with stable regulatory frameworks, faster consenting and dedicated export corridors. These measures would reduce the risk that European hydrogen demand grows faster than domestic offshore supply.

These quantitative findings can help European governments reassess offshore targets, export corridors and carrier choices as national hydrogen strategies are updated. The modelled west-to-east offshore hydrogen export pattern would reshape European energy flows by connecting Atlantic and North Sea offshore resources to continental demand centres. It can also strengthen the net-zero transition by improving supply diversification, reducing exposure to fossil-gas import dependence and contributing to the wider energy trilemma of decarbonisation, security and affordability.

4 Methods

4.1 Tech-economic model of offshore hydrogen production

The offshore wind generation and cost are assessed within the EEZs of the selected coastal European countries, i.e., Belgium, Denmark, France, Germany, Ireland, Netherlands, Norway, Portugal, Spain, Sweden, and the UK [30]. Aquaculture Sites, Special Areas of Conservation, and military areas were excluded from the available offshore hydrogen production areas, the geo-boundary of which is obtained from the European Marine Observation and Data Network (EMODnet) [19]. Within these geographical boundaries, vectorized wind velocity data, as well as other weather data that could affect wind generation (temperature and air pressure) are obtained from the ERA-5 product of the fifth-generation European Centre for Medium-Range Weather Forecasts (ECMWF) reanalysis for the global climate and weather in the last five years. Data are provided at 1 hour and $0.5^\circ \times 0.5^\circ$ temporal and spatial resolutions (equivalent to about 45km by 56 km in South Europe) for the last five years [16].

Wind power generation and capacity factors are calculated on an hourly basis using the power curve of an NREL reference 15 MW wind turbine, where the technical parameters are described in [17]. The power curve is described at 1m/s wind speed resolution. For fractional wind speed, linear interpolation is used to estimate the power of the wind turbine. Turbine layouts are optimised by balancing the power density and interference from wake effects. The wake effect of a single turbine is simulated using the Jensen-Gaussian model, where the effectiveness has been demonstrated in [31]. When characterising the cumulative wake effects from multiple upstream-located wind turbines, multiple wake effects could overlap. Previous studies usually consider a single layer of wake effect only or use enumeration-based methods, which will increase the computation burden exponentially with the number of mutually interfered wind turbines and massive historical data [32, 33]. To overcome this challenge, a recursive-tree-based method is developed, which in practice could reduce the simulation time by 79% with less than 1% sacrifice of accuracy [34]. The specific methodologies and parameter values are provided in the Supplementary Information.

The LCOH of the offshore wind farm is calculated using a bottom-up approach, including four cost modules, development expenditure (DEVEX), capital expenditure (CAPEX), operational expenditure (OPEX), and decommissioning expenditure (DECEX). These cost modules are further broken down into different terms and components in the Supplementary Information. Bespoke techno-economic parameters for Europe are obtained from Offshore Renewable Energy Catapult, including environmental survey cost, wind turbine cost and cable cost [35]. Generally, given all the

parameters of the model, it requires five additional inputs, i.e., water depth, capacity factor (as calculated above), distance to the nearest port and connecting point, and distance between wind turbines to generate the geolocation-based LCOH. Water depth (bathymetry data) within the EEZs of the studied countries is obtained from the bathymetry database published by EMODnet [18]. Data were converted and applied to grid cells with $0.001^\circ \times 0.001^\circ$ spatial resolution (equivalent to about 0.11 km by 0.06 km near the UK). The water depth affects the type of wind farms as well as the amount of material used for building the platform. For water depths less than 30m and between 30-60m, the potential of monopile structure and jacket structure (fixed-bottoms) wind turbines is explored. The deployment of floating turbines in deeper water environments (over 60m) is also evaluated, but would entail significantly higher costs both for installation and maintenance. For EEZs of studied countries, the North Sea areas are relatively shallow and can be deployed with fixed bottom wind turbines, while the water depths of Iberian Peninsula marine areas increase faster with the distance to the coast. The minimum distance to the port affects the installation cost when shipping materials to offshore locations, as well as the annual maintenance expenses. The major port and city locations in the studied countries are obtained from EMODnet [19]. The minimum distance to connecting points determines the capital cost of export cables that transmit electricity generation from offshore to onshore substations. Distances between wind turbines are determined based on the principle that the maximum interference from adjacent wind turbines from the wake effect should be less than 5%. It affects the cost of array cables that collect the generations from individual wind turbines and converge on the offshore substation.

In the offshore electrolysis platform, there are several components consuming the energy from offshore wind, including water pumps and desalination, electrolyzers, hydrogen compressors, and standby batteries. The techno-economic characteristics of the Alkaline Electrolyser (AEC) are summarised in Supplementary Table 3. The AEC system is currently the most mature and durable among these three options, and therefore is studied in this paper. The data on the capital cost and electrical efficiency were obtained from projections by the Offshore Renewable Energy Catapult and the UK's National Hydrogen Strategy. The efficiencies of AEC are projected to increase from 0.79 in 2030 to 0.82 in 2050 [36]. The capital cost includes previously mentioned key components, as well as necessary auxiliary systems such as platforms and pipelines. Annual OPEX includes maintenance, stack replacement, and labour costs are expressed as a percentage of initial capital investments. Due to the fluctuations of offshore wind and mismatch with the electrolyser's operating point, the electrical and hydrogen storage facilities are essential. Water pumps, desalination, and hydrogen compressors should be sized based on hydrogen production. Therefore, their total power consumption is dependent on the hydrogen production capacity and adds up to the rated power of the wind generation with a certain margin. On this basis, a set of algebra equations (Supplementary Information, Section 3.3) is formulated based on the maximisation of utilisation rate to determine the optimal size of the electrolyser and other accessories. The operating technical parameters and various costs in the future are projected by [35]. By accounting for the DEVEX, CAPEX, OPEX, and DECEX of these devices, the LCOH can be calculated.

In addition to the LCOH map, to obtain the cost-supply curves of hydrogen, the available supplied quantity of hydrogen at each level of LCOH should be further evaluated. The potential installed offshore wind capacity for each grid cell was calculated by multiplying the turbine’s rated power by the number of turbines that could fit maximally into an individual cell. Distances between wind turbines are determined as aforementioned to control the wake effect below the threshold while maximising the power density. The vessel density could affect the power density given a specific layout of a wind farm, which is obtained from EMODnet with 1 km resolution [37]. It is assumed that the power density could decrease linearly with the increase of vessel density up to 50% [38]. In this study, we consider three future scenarios for LCOH reductions (high, median, and low) based on expert elicitation surveys [21]. Compared with 2020, the reduction rates in 2030, 2040, and 2050 in the example of median scenarios are projected at 21%, 35%, and 49% for fixed wind farms, and 2%, 17%, and 40% for floating wind farms, respectively.

4.2 Domestic offshore wind and hydrogen integration model

The offshore power generation is integrated into the domestic power system via marine cables, while for bulk utilisation, the hydrogen production is connected to the domestic national natural gas system via pipelines through hydrogen blending techniques (assumed from 0% to 100% from now to 2050). Hydrogen blending refers to injecting hydrogen into existing natural gas pipelines, being fully mixed, and transported to other locations for further use. Compared with local hydrogen utilisation or dedicated hydrogen pipelines, hydrogen blending can provide a cost-efficient interim solution to rapidly promote the scale of hydrogen demand. Hydrogen blending has received substantial support and has been successfully demonstrated in numerous countries, including the European hydrogen backbone [39], HyDeploy in the UK [40], and HyBlend in the US [41]. The UK’s Department for Energy Security and Net Zero permitted distribution-level hydrogen blending in December 2023, and is actively seeking more evidence to extend it to transmission level [42]. According to the Energy Network Association, the UK aims to achieve 100% hydrogen networks by 2050 [43]. Therefore, the utilisation and integration of offshore power and hydrogen production is constrained by the national energy systems, in terms of demands, network transmission capacities, and operation schemes. To evaluate the potential of the domestic energy system for accommodating offshore resources, especially for fully exploring island countries like Ireland with abundant offshore resources and inadequate domestic energy demands, a coordinated optimal operation model (COM) for the national energy system, including both power and gas networks, is developed. The optimisation model will be conducted on an hourly basis throughout the year, simulating the optimal system operating state with the lowest operation cost while maximising the offshore resource integration.

The power system module in the COM constitutes a unit commitment framework to schedule the resources in the day ahead for balancing the variation of energy demands and uncertain renewable generation on the nodal scale. The unit commitment cost includes fuel costs and start-up costs for thermal units. For the All Ireland Island power system, the practical operational policy roadmap from 2023-2030, as outlined by

EirGrid and Soni, is incorporated in the model, such as Inertia constraints, Minimum Number of Conventional Units, System Non-Synchronous Penetration (SNSP) [44]. The flexibility constraints for generators are incorporated, including unit ramping limits, minimum online-offline times, minimum output levels and must-run constraints. The power network congestion, which may also hinder the integration of offshore wind generation, is modelled using the DC power flow model. The dynamic rating of power lines with temperature variations is considered in summer and winter scenarios. Future construction plans for interconnections are considered, such as the 700 MW Celtic Interconnector with France in 2027 [45].

The natural gas system module is constructed to minimise the operating cost for gas systems while accommodating the hydrogen injection from offshore hydrogen. With the stochastic hydrogen injection, the gas compositions and corresponding physical properties (such as heat value and specific gravity) of the resulting gas mixtures will be uncertain, which can bring threats to the gas security and challenges to the optimal operation of gas systems. To overcome these issues, we first defined a gas security region according to the recently updated The Gas Safety (Management) (Amendment) Regulations 2023, as proved by the UK’s Legislation [46], which uses relative density, Wobbe index, and additionally Weaver flame speed factor, to limit the hydrogen fraction. The weymouth equation is used to describe the relationship between gas flow rate and pressure drops along a gas pipeline to avoid over-/under pressure issues in the network. However, due to the variant gas physical properties, the Weymouth equation becomes more nonconvex, which could lead to local optima and unstable solutions. Here we developed a reformulation method and solution algorithm, comprised of bespoke McCormick envelope, second-order cone reformulation, and sequential convex programming. Then, the optimisation problem can be robustly and efficiently solved by off-the-shelf commercial solvers (such as Gurobi). Compared with general nonlinear solvers (such as IPOPT), the computation efficiency can be improved by over 90% [47]. When multiple pipelines intersect, different compositions of gas will be mixed evenly at the joint point. This process is also incorporated in our model. Identically, this process will generate bilinear constraints and cause computation burdens, which is also addressed by our previously introduced solution framework (mathematical models can be found in Section 4 of Supporting Information).

For an accurate description of power and gas system operations, high temporal-spatial precision and long-span power generation, consumption, and transmission datasets are employed. On the generation side, the detailed operational parameters of All-Ireland Island generators, such as capacity, and piecewise heat rate curves, are obtained from the Single-Electricity-Market (SEM) Committee, which has been used and validated in EirGrid’s publications [48]. For non-gas fossil fuel units such as coal and oil, the commodity cost and heat value are also counted based on the SEM model. For gas-fired units, the fuel cost is counted on the gas system side, and the heat value is modelled dynamically with the variation of hydrogen fractions. For onshore wind and solar generations, the wind speed and solar radiation are obtained from ERA-5, and the Beckman model is used to calculate the output of solar generations. The hydro generation data are obtained from historical operations [49]. For the consumption sector, the trends of gas demand in the future scenarios are predicted according to the Gas

Forecast statement by Gas Network Ireland [50], with various cross-sector considerations, such as the decommissioning of coal-fired power plants (e.g., Moneypoint units), 80% renewable generation ambition from Irish government’s Climate Action Plan, and impacts of banned new gas boiler connections and Gross Domestic Product (GDP) on residential, industrial and commercial demands, construction of new compressed natural gas stations. As a result, the gas demand will increase in the short term, reach a peak around 2025, and then continue to decrease. Similarly, the daily electricity peak demands from now to 2032 are predicted by a bottom-up approach in EirGrid’s Ten-Year Generation Capacity Statement [51]. It covers the forecasting values from now to 2032 in typical seasonal scenarios (winter, summer, and spring-autumn). Considering the increasing trend of annual power demand is steady, the Autoregressive Integrated Moving Average (ARIMA) is used to fill the missing values between 2033 and 2050. 15-minute-resolution electricity and gas demand curves are used for daily operational analysis. They are obtained from EirGrid’s smart grid dashboard and the Gas Network Ireland data dashboard in different seasonal scenarios [52, 53]. For the power transmission system, high voltage power lines over 110 kV are modelled, and the network topology, geographical locations, and electrical parameters (such as impedance) are obtained from EirGrid’s Transmission Forecast Statement [54]. The high-pressure gas transmission pipeline network (above 16 bar) is modelled, and similarly, the network topology, geographical locations, and pipeline length are obtained from Gas Network Ireland’s Network Development Plan [55]. Friction factors and diameters are assumed according to a dedicated case study of Ireland [56]. The capacities of power lines are assumed to grow annually at the same rate as the demand growth.

4.3 International green hydrogen trading model

After domestic consumption, surplus hydrogen and ammonia can be transported by ship to demand centres elsewhere in Europe. We formulate an international hydrogen trading model that uses the country-level cost-supply curves and available export potentials derived above to simulate hydrogen and ammonia flows among the studied European countries in 2030, 2040 and 2050. On the supply side, countries with published offshore wind targets, such as Ireland [57] and the UK [58], use those targets as maximum offshore capacities. For countries without complete year-specific targets, we use tentative goals from the European Union’s non-binding agreement on offshore wind deployment [59]. Hydrogen and ammonia demands are taken from the European Hydrogen Backbone demand projection [60].

The integrated model minimises total production, shipping and external-import costs subject to national supply capacities, domestic export limits, country-level hydrogen and ammonia demand, shipping-route capacities and material balances (Supplementary Information, Section 5). Hydrogen and ammonia production costs are represented through piecewise-linear interpretations of the cost-supply curves. Shipping costs include annualised ship investment costs and operating fuel costs, with route distances derived from port-to-port shipping paths [19]. External imports are represented by an international import node so that unmet European demand can be supplied from outside the modelled offshore system when domestic offshore supply is insufficient or when the outside option is cost competitive. Because the supply curves

and ship investment decisions introduce discontinuities, the model is formulated as a mixed-integer linear programming problem and solved in MATLAB using Gurobi.

For the lowest-LCOH-only counterfactual, total European hydrogen and ammonia demand is allocated to offshore supply segments in ascending LCOH order. This baseline uses planned offshore capacity and capacity factors to limit country supply, but deliberately excludes domestic absorption limits, shipping costs, demand geography and the competitive external-import price. It is therefore used only as a diagnostic baseline for testing whether exporter rankings can be inferred from production cost alone.

For the outside-option sensitivity, the integrated trade model is solved repeatedly with a delivered hydrogen-equivalent external import price from 2 to 6 EUR kg⁻¹ H₂-equivalent, corresponding to 60–180 EUR MWh⁻¹. Hydrogen and ammonia imports are assigned the same hydrogen-equivalent delivered price so that the scan isolates the threshold at which European offshore supply becomes competitive with non-European low-carbon imports.

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Author contributions

S.W. conceived the study, developed the modelling framework and MATLAB code, performed the analysis, prepared the figures and wrote the original manuscript. M.M.B. contributed to the energy-system interpretation, policy context and manuscript review and editing. Both authors reviewed and approved the final manuscript.

Competing interests

The authors declare no competing interests.

Data availability

The public data sources used in this study are cited in the References and described in the Methods and Supplementary Information. Derived tables and figure inputs generated by the analysis are available in the project repository. Large raw datasets and intermediate MATLAB checkpoint files are not stored in the repository because of file-size constraints, but they can be regenerated from the cited public sources and the provided code.

Code availability

MATLAB code used in this study is available on GitHub at <https://github.com/ShengWang-EE/HyExport>.

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